Latest Trends in Cloud and DevOps



Latest Trends in Cloud



Engage and Think



A large company recently integrated a new AI-powered analytics platform to improve decision-making. Despite the advanced features and significant investment, most team members continue to use traditional spreadsheets and manual reporting. As a result, the adoption rate is low, and the platform's potential remains untapped.

Why do you think professionals often resist using new technologies, even when they offer better features and results?

Learning Objectives

By the end of this lesson, you will be able to:

- Identify emerging trends in cloud computing to determine their business relevance
- Compare cloud models and technologies to select the best fit for modern workloads
- Evaluate industry use cases to assess how organizations adopt cloud innovations for growth
- Examine future-facing solutions like AI-as-a-Service (AIaaS) and quantum computing to prepare for upcoming disruptions



Cloud Computing Evolution and Market Landscape

What Is Cloud Computing?

It delivers IT (Information Technology) resources on demand over the internet with pay-as-you-go pricing. Users can access services like storage, databases, and software without managing physical servers.



Cloud computing empowers organizations to scale faster, reduce costs, and focus on innovation instead of infrastructure management.

Cloud Computing Evolution Timeline: Overview

Cloud computing has evolved through major phases to become the backbone of modern IT. The timeline below highlights key milestones shaping today's cloud ecosystem:

1950s–1970s	**Mainframe era:** Delivering centralized computing with limited user access
1980s–1990s	**Personal computing era:** Enabling decentralized computing with individual workstations
Late 1990s	**Virtualization era:** Running multiple virtual machines on a single physical server
2000s	**Public cloud era:** Providing scalable resources over the internet
2010s–Present	**Modern cloud era:** Supporting hybrid, multi-cloud, and serverless architectures

Mainframes to Virtualization Era (1950s–1990s)

Below are the key phases in computing history that laid the foundation for modern cloud infrastructure, starting with centralized mainframes and moving to virtualization for efficient resource use:

Mainframe computing

Centralized processing with limited user access and high operational costs

Personal computing

Decentralized computing through affordable personal computers for individual users

Virtualization technology

Creation of virtual environments on a single machine to enhance hardware efficiency and flexibility

Birth of the Public Cloud Era (2006–2010s)

The introduction of public cloud services transformed how organizations deploy and manage applications, offering unprecedented scalability and cost-efficiency.

1

Launch of AWS
Amazon Web Services introduced EC2 and S3 in 2006

2

Launch of GCP
Google Cloud Platform was launched in 2008

3

Launch of Azure
Microsoft Azure was introduced in 2010

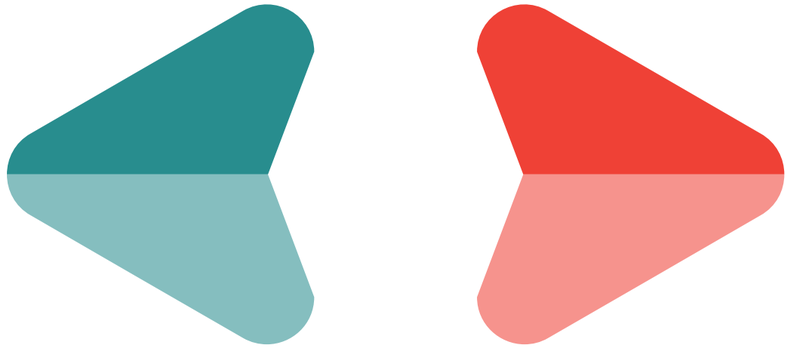
4

Adoption drivers
Adoption driven by cost savings, scalability, and recovery

Modern Cloud Era Capabilities

Contemporary cloud platforms offer advanced features that empower organizations to innovate and respond swiftly to market changes.

Auto-scaling
Dynamically adjusts resources based on demand to maintain performance and cost-efficiency



Serverless computing
Executes code in response to events without the need to manage or provision servers

Global Cloud Market Growth (2024–2028)

The following projections highlight the rapid expansion of the global cloud market, fueled by digital transformation, AI integration, and hybrid workforces:



Global cloud computing market projected to grow from $678.8 billion in 2024 to over $1.44 trillion by 2028

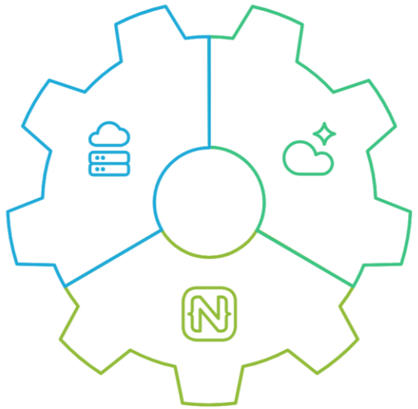


CAGR of approximately 16.4% over the forecast period

Key Growth Drivers

As global cloud spending accelerates, the following drivers explain what is fueling this growth across industries:

Adoption of hybrid and multi-cloud



Increase in AI/ML workloads

Rise of cloud-native development

Industry-Specific Cloud Applications

Cloud computing is changing how industries operate by providing flexible solutions that enhance agility, scalability, and decision-making. Here are some key industries that use cloud technology:

Healthcare
Secures patient data and enables remote diagnostics

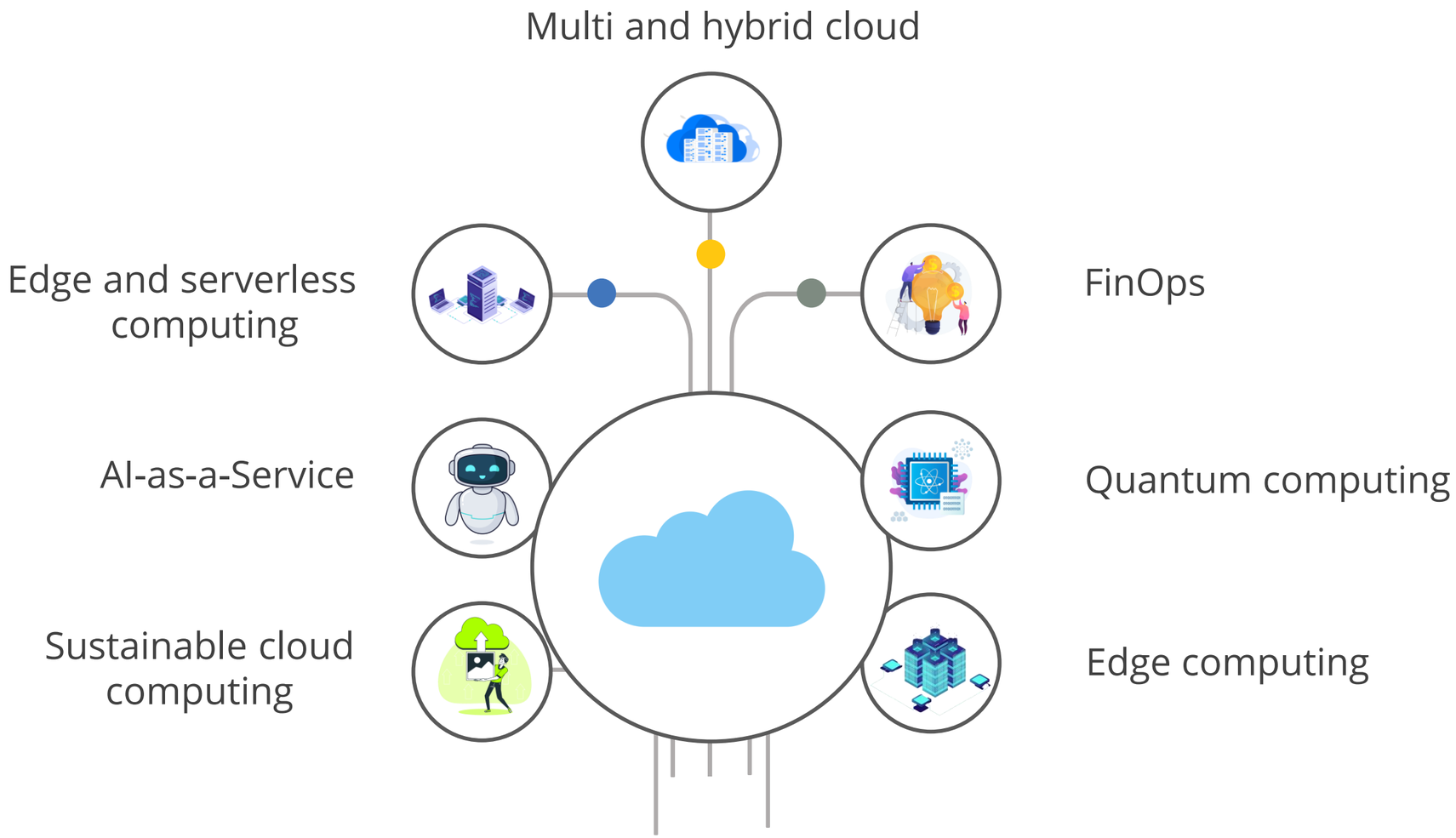


Retail and e-commerce
Optimize inventory and handle traffic spikes

Banking and finance
Secure transactions and ensure compliance

What Are Latest Trends in Cloud?

Cloud computing continues to evolve, driven by innovations that enhance scalability, flexibility, and intelligent automation. The following are key emerging trends shaping the future of cloud computing:



Multi-Cloud and Hybrid Cloud Models

Introduction to Multi-Cloud

It is the use of two or more public, private, or hybrid cloud providers to run applications instead of relying on a single provider.



Multi-cloud strategies offer greater flexibility, improved resilience, and help avoid vendor lock-in by leveraging multiple cloud providers.

Multi-Cloud: Benefits

The following are the advantages of using a multi-cloud approach in applications:

Combines services from multiple providers

Enhances redundancy, resilience, and availability



Offers flexibility in features and pricing

Minimizes vendor lock-in and supports compliance

Challenges of Multi-Cloud

Below are the common challenges organizations face when managing a multi-cloud environment:

Security gaps
Enforcing consistent policies and access controls

Integration challenges
Syncing data across environments

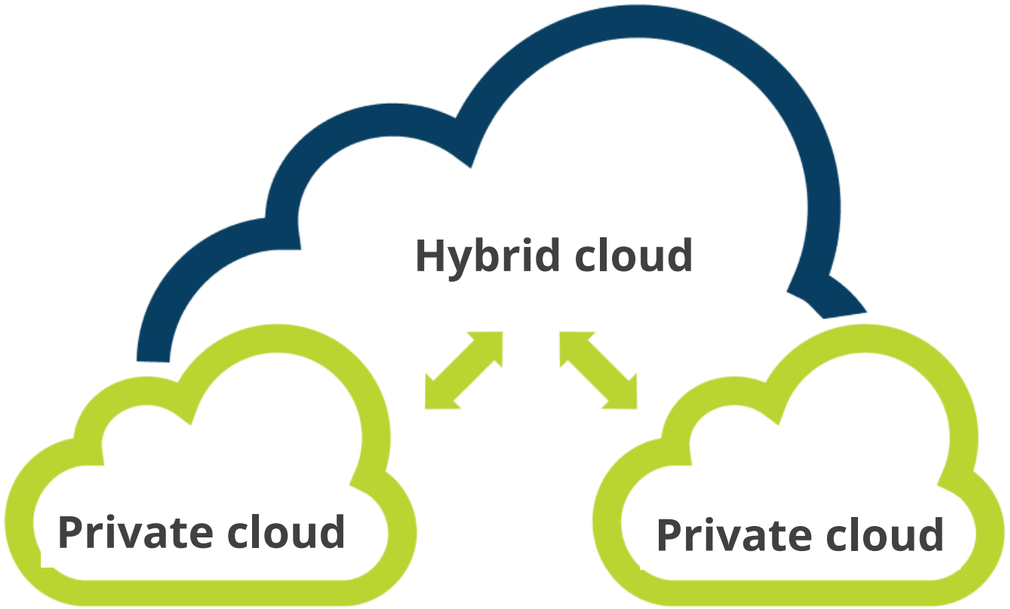


Increased complexity
Managing multiple interfaces and billing systems

Operational overhead
Needing skilled teams and monitoring tools

Introduction to Hybrid Cloud

It is a computing environment that combines on-premises infrastructure (or private clouds) with public cloud services, allowing data and applications to move between them.



This approach offers greater flexibility, optimizes workload distribution, and supports regulatory or latency requirements.

Hybrid Cloud: Benefits

Below are the key benefits that make hybrid cloud appealing for enterprise workloads and compliance-heavy environments:

Scalability
Expanding resources as needed

Security and compliance
Keeping data secure and regulatory compliant



Legacy integration
Connecting older systems with new cloud apps

Cost efficiency
Using a public cloud only when needed

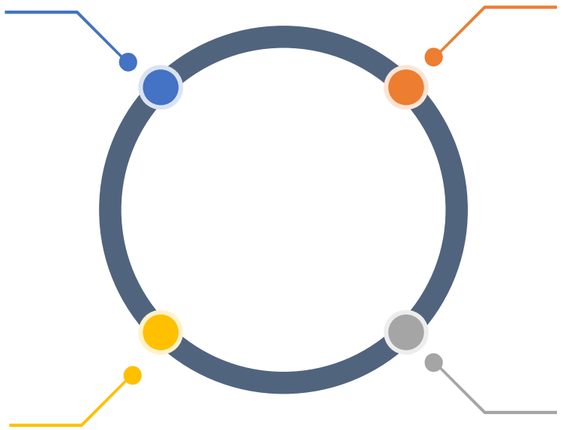
Business continuity
Ensuring operations with minimal downtime

Challenges of Hybrid Cloud

Organizations commonly face the following challenges when implementing and managing hybrid cloud environments:

Integration complexity
Sync on-premises and cloud systems

Security consistency
Maintain unified access and protection



Cost management
Track usage and avoid hidden costs

Operational overhead
Manage and troubleshoot hybrid setups

Hybrid vs. Multi-Cloud

Below is a comparison of hybrid and multi-cloud approaches to help determine which strategy best fits specific business needs:

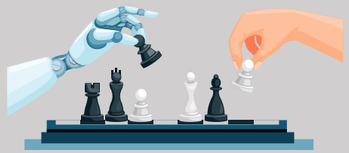
Criteria	Hybrid cloud	Multi-cloud
Deployment	Combines private and public cloud	Uses multiple public cloud providers
Purpose	Enhances control and compliance	Improves flexibility and avoids lock-in
Control	Manages both on-premises and cloud systems	Relies on external cloud platforms

Hybrid vs. Multi-Cloud

Below is a comparison of hybrid and multi-cloud approaches to help determine which strategy best fits specific business needs:

Criteria	Hybrid cloud	Multi-cloud
Complexity	Requires system integration	Requires provider coordination
Security	Centralized governance	Distributed security policies
Use case	Regulated industries with legacy systems	Global businesses with diverse workloads

Use Case: Capital One's Cloud Transformation



Challenge



Solution



Outcome

Capital One aimed to enhance its operational efficiency, scalability, and customer experience. The company faced challenges with its on-premises data centers, including high operational costs, limited scalability, and the need for robust security and compliance measures. Additionally, Capital One sought to accelerate its digital transformation to remain competitive in the rapidly evolving financial services industry.

Source: https://www.thinklogic.com/post/cloud-migration-case-study-how-capital-one-successfully-transitioned-to-the-cloud

Use Case: Capital One's Cloud Transformation



Challenge



Solution



Outcome

- Adopted a cloud-first strategy by migrating to AWS for flexibility and scale
- Modernized key applications using cloud-native architectures
- Strengthened security and compliance to meet financial regulations
- Enabled Agile development through CI/CD and DevOps practices

Source: https://www.thinklogic.com/post/cloud-migration-case-study-how-capital-one-successfully-transitioned-to-the-cloud

Use Case: Capital One's Cloud Transformation



Challenge



Solution



Outcome

- Reduced operational costs by decommissioning data centers
- Gained global scalability for real-time financial services
- Improved security posture across cloud workloads
- Accelerated feature delivery for better customer experience

Source: https://www.thinklogic.com/post/cloud-migration-case-study-how-capital-one-successfully-transitioned-to-the-cloud

Quick Check



A global company uses AWS for customer applications and Azure for internal analytics. They want flexibility to avoid vendor lock-in and optimize cost-performance. What cloud approach are they using?

A. Private cloud

B. Hybrid cloud

C. Multi-cloud

D. Public cloud

Serverless Computing

What Is Serverless Computing?

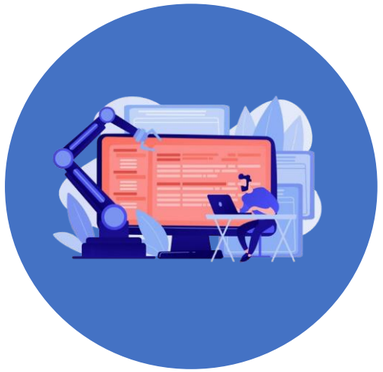
It is an application development model that allows developers to build and run code without managing servers or backend infrastructure, allocating machine resources on demand.



Serverless computing is managed by a cloud service provider (CSP) that offers a platform, tools, and services that streamline the development process.

Why Use Serverless Computing?

Serverless computing lowers overhead and costs by automating infrastructure management and scaling, allowing developers to focus on building applications instead of managing servers. Benefits include:



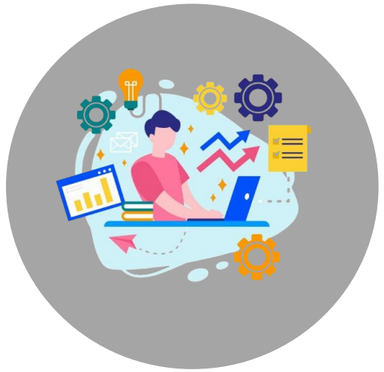
Automation

It removes the toil of managing servers by automating tasks.



Scalability

It scales up and down automatically in response to traffic.



Productivity

It empowers developers to focus on writing code and optimizing business logic.

When to Use Serverless Computing Model?

Serverless architecture offers multiple use cases across various industries, such as:



When to Use Serverless Computing Model?

Media processing

Automatically scales for tasks like encoding, resizing, and filtering media files

Building APIs

Eases API development with auto-scaling and no infrastructure management

Webhooks

Triggers functions in response to events for real-time processing

When to Use Serverless Computing Model?

Microservices

Enables independent deployment and scaling of modular services without server management

Chatbots

Scales the application to handle real-time user interactions and traffic variations efficiently

Event-driven applications

Executes functions based on specific triggers or events without always-on servers

Working of Serverless Computing Model

1. Function creation:
Developers create functions to handle application tasks.



2. Event definition:
They define events to trigger functions.



3. Event invocation:
User actions trigger events that are sent to the cloud provider.



4. Resource allocation:
The cloud provider dynamically allocates resources to handle the event.



5. Function execution:
The function processes the event and returns the appropriate response to the user.



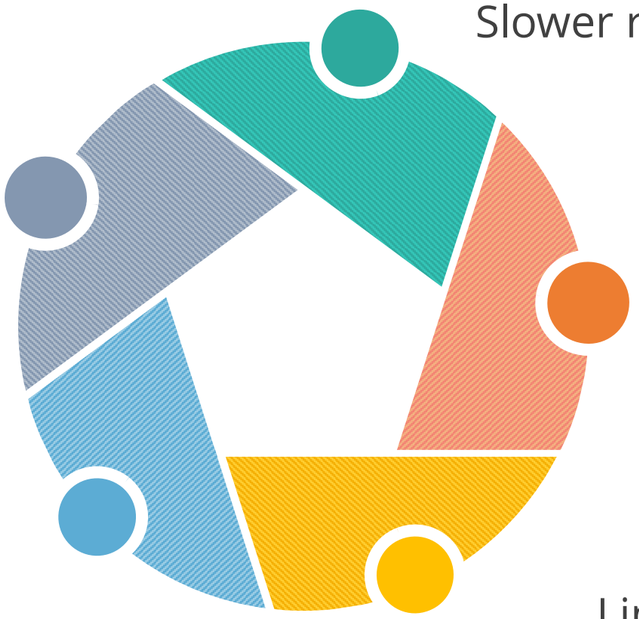
After the function is completed, the resources are deallocated, and no intermediate state or resource is maintained unless necessary.

Limitations of Serverless Computing

Below are the key limitations to consider when planning a serverless architecture:

Cold start delays
Slower response when functions are idle

Monitoring challenges
Harder to trace issues in distributed systems



Limited execution time
Runtime limits (example: 15 minutes on AWS Lambda)

Not for long-running tasks
Best for short and stateless functions

Vendor lock-in
Limited portability across platforms

Applications Based on Serverless Architecture

The following are the real-world examples of serverless applications:



Coca Cola

Leverages serverless architecture for Freestyle vending machines, scaling to manage 80 million monthly transactions and cutting costs



Equinox Media

Uses AWS serverless architecture to enable scalable user experiences and prevent performance bottlenecks

Applications Based on Serverless Architecture

The following are the real-world examples of serverless applications:



BMW

Utilizes AWS serverless services to manage and analyze vast data efficiently



Autodesk

Leverages AWS serverless technologies to reduce costs and accelerate custom account setups

Top Providers of Serverless Architecture

The following are the top providers of serverless applications:



AWS



Microsoft Azure



Vercel



Netlify



Google Cloud



Cloudflare

Edge Computing

What Is Edge Computing?

It is a distributed information technology (IT) architecture in which client data is processed at the periphery of the network, as close to the originating source as possible.



Edge computing reduces latency, improves response times, and supports real-time decision-making in applications like IoT and smart devices.

Benefits of Edge Computing

Below are the key benefits of edge computing that make it ideal for latency-sensitive and bandwidth-intensive applications:

Operational continuity
Maintains function during cloud connectivity issues

Reduced latency
Enables real-time responses by processing data locally

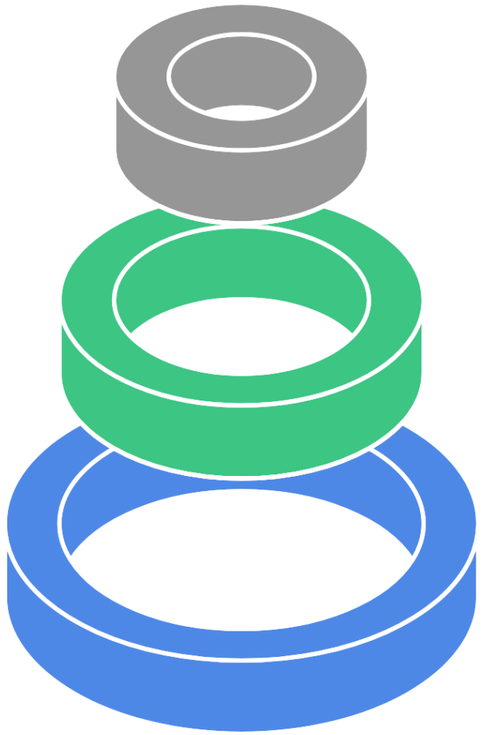


Improved security
Processes sensitive data before sending it to the cloud

Bandwidth efficiency
Lowers network load by minimizing cloud data transfer

Edge Computing Architecture

It processes data near its source (example: sensors or devices) instead of relying solely on centralized cloud servers. The following is a breakdown of the architecture:



Cloud
Store and analyze selected data centrally

Edge nodes
Process and filter data locally

Edge devices
Generate data at the source

Edge Computing: Use Cases

Below are the real-world scenarios where edge computing is used to enhance performance and efficiency:



Autonomous vehicles
Analyze traffic, surroundings, and make real-time decisions

Smart cities
Manage local control of lighting, traffic, and surveillance

Industrial IoT
Detect faults and trigger automated actions on factory floors

Healthcare devices
Monitor vital signs and trigger alerts for anomalies

Edge Computing: Challenges

Below are the key challenges organizations face when deploying and managing edge computing environments:

Device management
Monitor and update devices across large networks



Data security
Protect sensitive data at edge nodes and gateways

Limited resources
Manage low power, memory, and processing capacity

Latency variability
Manage inconsistent response times across edge networks

Quick Check



A manufacturing company installs smart sensors on its factory machines. These sensors detect anomalies and immediately stop production without relying on cloud servers. What computing model best supports this?

A. Cloud computing

B. Grid computing

C. Edge computing

D. Client-server model

Sustainable Cloud Computing

What Is Sustainable Cloud Computing?

It refers to the practice of designing, operating, and managing cloud computing environments in a way that minimizes environmental impact, primarily by reducing energy consumption and carbon emissions.

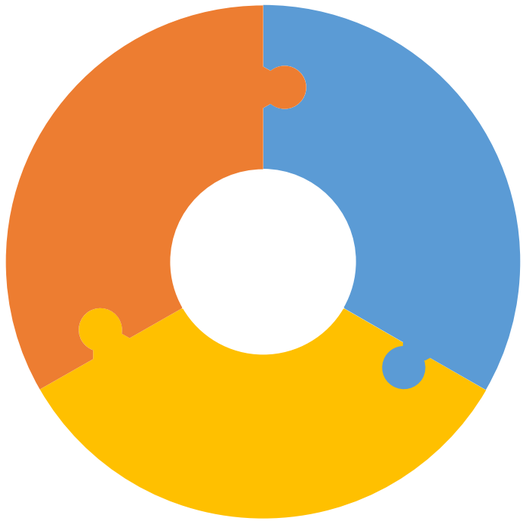


It blends environmental responsibility with technology efficiency, aiming to deliver digital services while being conscious of ecological footprints.

Benefits of Sustainable Cloud

Below are the major benefits organizations achieved by moving toward more sustainable cloud operations:

Lower costs
Reflects operational savings from efficient energy use



Fewer emissions
Indicates reduced carbon footprint through renewable energy

Stronger reputation
Points to brand value gained through sustainability efforts

Challenges in Adopting Sustainable Cloud

Below are some common challenges organizations face when implementing sustainable cloud practices:

High costs
Requires significant upfront investment in green technologies and migration



Low transparency
Lacks clear verification of sustainability claims by providers

Limited access
Lacks availability of sustainable infrastructure in certain regions

Performance trade-offs
Impacts flexibility and performance due to optimization efforts

Cloud Provider Sustainability Initiatives

The following are the sustainability goals and achievements of leading cloud providers working to make cloud infrastructure more eco-friendly:

Google Cloud

Aims for 24/7 carbon-free energy by 2030

Microsoft Azure

Pledges to be carbon negative by 2030 and remove historical emissions by 2050

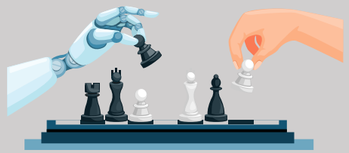
Amazon Web Services

Plans to use 100% renewable energy by 2025

IBM Cloud

Focuses on AI-driven efficiency and achieving net-zero emissions by 2030

Case Study 1: Google Cloud’s 24/7 Carbon-Free Energy Goal



Challenge



Solution



Outcome

Google aimed to lead the way in sustainable cloud computing by 2030 by operating entirely on 24/7 carbon-free energy. As demand for AI and digital services increased, so did energy consumption across its data centers. The challenge was to meet global performance needs while eliminating reliance on carbon-based power and operating cleanly in every location, at every hour.

Source: https://sustainability.google/reports/247-carbon-free-energy/

Case Study 1: Google Cloud’s 24/7 Carbon-Free Energy Goal



Challenge



Solution



Outcome

- Adopted a carbon-free energy model, matching energy use to local renewable supply
- Invested in solar, wind, and geothermal projects across multiple regions
- Launched the Carbon-Free Energy (CFE) Manager to automate and optimize clean energy usage
- Piloted nuclear energy partnerships (example: Kairos Power) for AI data centers

Source: https://sustainability.google/reports/247-carbon-free-energy/

Case Study 1: Google Cloud’s 24/7 Carbon-Free Energy Goal



Challenge



Solution

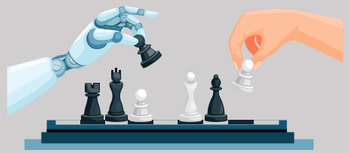


Outcome

- Achieved 90% carbon-free operations in several global data centers
- Accelerated clean energy innovation with real-time tracking systems
- Set an industry standard for hour-by-hour renewable energy adoption
- Inspired other cloud providers and governments to pursue similar goals

Source: https://sustainability.google/reports/247-carbon-free-energy/

Case Study 2: Microsoft Azure’s Carbon Negative Commitment



Challenge



Solution



Outcome

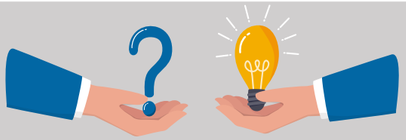
Microsoft pledged to become carbon negative by 2030 and eliminate all historical carbon emissions by 2050. The main challenges included curbing emissions from its expanding cloud infrastructure, adhering to international compliance standards, and encouraging suppliers to reduce their carbon footprint, all while continuing to grow its operations globally.

Source: https://blogs.microsoft.com/blog/2020/01/16/microsoft-will-be-carbon-negative-by-2030/

Case Study 2: Microsoft Azure’s Carbon Negative Commitment



Challenge



Solution



Outcome

- Introduced an internal carbon fee to fund sustainability projects across departments
- Invested in over 10.5 gigawatts of renewable energy capacity globally
- Launched tools to help customers track cloud carbon emissions
- Engaged suppliers to adopt 100% carbon-free energy by 2030

Source: https://blogs.microsoft.com/blog/2020/01/16/microsoft-will-be-carbon-negative-by-2030/

Case Study 2: Microsoft Azure’s Carbon Negative Commitment



Challenge



Solution



Outcome

- Reduced overall emissions and increased renewable energy procurement
- Enhanced transparency with annual Sustainability Reports
- Strengthened partnerships with governments and NGOs for climate action
- Positioned Azure as a preferred platform for ESG-conscious organizations

Source: https://blogs.microsoft.com/blog/2020/01/16/microsoft-will-be-carbon-negative-by-2030/

Quick Check



An enterprise wants to reduce its carbon footprint by migrating to cloud services powered by renewable energy. Which cloud sustainability initiative best supports this goal?

A. Moving to a private data center

B. Using only cloud services with autoscaling

C. Choosing providers with 24/7 carbon-free energy goals

D. Encrypting all traffic between cloud regions

AI-as-a-Service (AIaaS)

What Is AI-as-a-Service (AIaaS)?

It is a cloud-based offering that allows individuals and businesses to access artificial intelligence (AI) tools and capabilities without needing to build or maintain their own AI infrastructure or expertise.

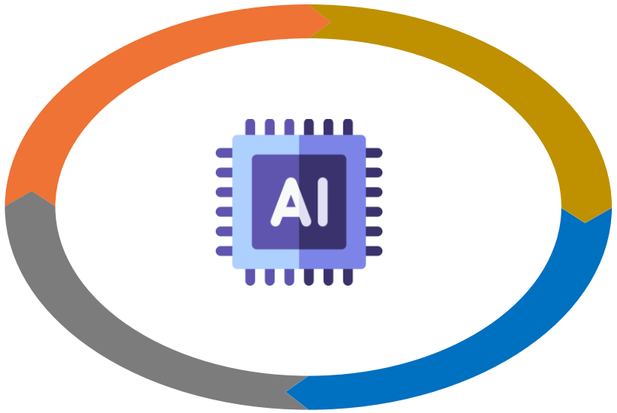


Core Components of AIaaS

Below are the core services and tools typically offered within an AIaaS platform:

Pre-trained models
Used for tasks like vision, language, and translation

Machine learning platforms
Used to build, train, and deploy custom models



AI APIs and SDKs
Used to integrate AI into applications

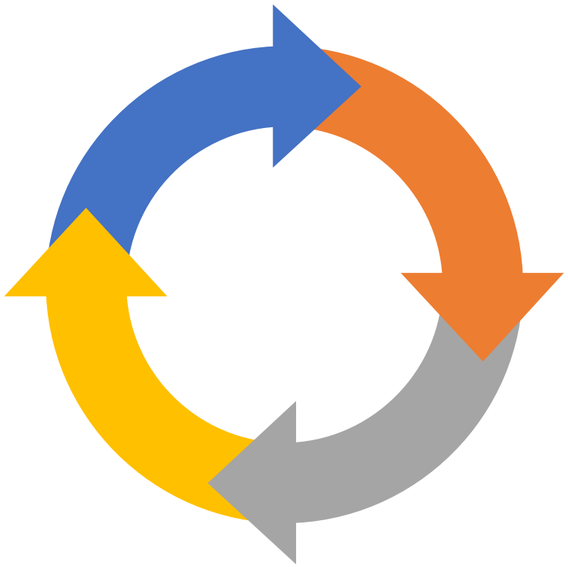
AI infrastructure
Used to run models with scalable compute

Benefits of AI-as-a-Service

Below are the key advantages organizations gain by adopting AIaaS solutions:

Faster time to market
Launch with ready-made models

Scalability
Scale workloads as demand grows



Cost efficiency
Skip upfront AI infrastructure and hiring costs

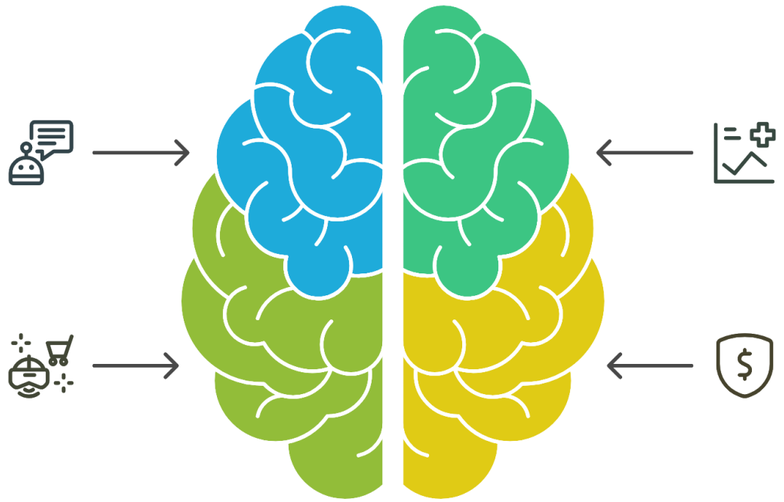
Advanced AI access
Use enterprise-grade tools from top tech firms

Use Cases of AI-as-a-Service

Below are some common industry applications where AIaaS is helping solve real-world challenges:

Customer support
Automate queries with chatbots and voice assistants

Healthcare
Predictive analytics for diagnosis and planning



Retail
Product recommendations from user behavior

Finance
Fraud detection, credit scoring, and risk analysis

Major AIaaS Providers

Below are the major cloud platforms delivering enterprise-grade AI-as-a-Service solutions:



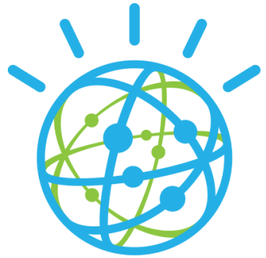
AWS AI/ML Services



Microsoft Azure AI



Google Cloud AI



IBM Watson

Quick Check



A startup wants to launch a chatbot without building its own AI models. They want a plug-and-play solution using existing APIs. Which option should they choose?

A. Quantum computing

B. AI-as-a-Service

C. Blockchain-as-a-Service

D. Edge analytics

Quantum Computing in the Cloud

What Is Quantum Computing?

It is a powerful new type of computing that uses tiny building blocks called qubits, which can perform many calculations simultaneously by following the strange rules of nature.



This makes it possible to solve very difficult problems, such as finding the best solution among many options or simulating how nature works, much faster than regular computers can.

Four Key Principles of Quantum Computing

Below are the four foundational principles that govern how quantum systems behave and enable quantum computing:

Entanglement
Qubits share states, even across distances

Decoherence
Qubits lose quantum state due to environment



Superposition
Qubits exist in multiple states at once

Interference
Qubit states combine to refine outcomes

Advantages of Quantum Computing

Below are the key areas where quantum computing offers potential breakthroughs beyond the capabilities of classical systems:

Speed and efficiency
Solves problems faster than classical systems

Complex problem solving
Handles intensive optimization and modeling

Cryptography
Secures or breaks encryption using quantum methods



Data analysis
Extracts insights from complex datasets

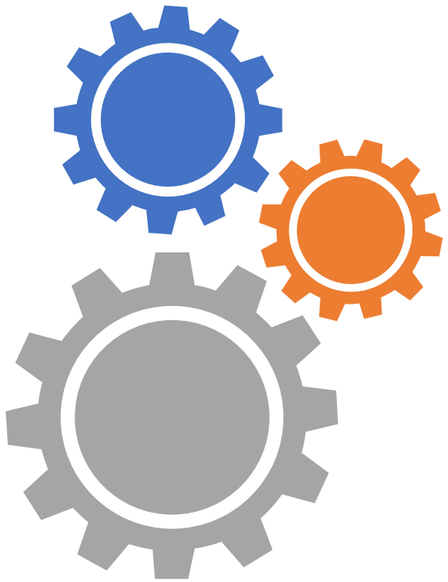
Scientific research Simulates physical and chemical systems

Artificial intelligence
Speeds up training and pattern detection

Potential Applications of Quantum Computing

Below are key domains where quantum computing is expected to deliver significant impact by solving complex problems more efficiently than classical systems:

Cryptography
Secures or breaks encryption to protect data



Optimization
Solves complex problems in logistics and finance

Material Science
Discovers new materials through atomic simulations

FinOps

What Is FinOps?

It is a cloud financial management practice that helps organizations manage, optimize, and control their cloud spending.



It's a blend of finance, engineering, and business teams working together to make data-driven decisions about cloud usage and costs.

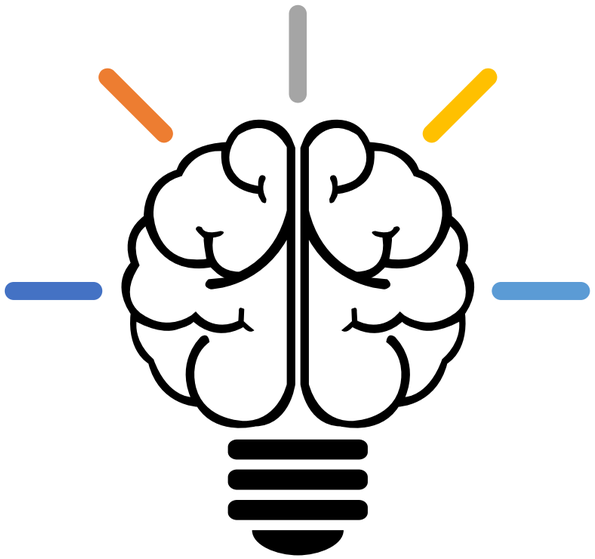
Key Principles of FinOps

Below are the guiding principles that help organizations build effective FinOps practices:

Team ownership
Teams manage their own usage and budgets

Real-time visibility
Cloud costs are tracked in near real time

Value-driven decisions
Cost trade-offs are evaluated for business outcomes



Shared accountability
Finance, engineering, and operations teams collaborate with each other

Uncompromised agility
Financial controls support innovation without slowing it down

FinOps Tools

Below are the tools used in FinOps to monitor, analyze, and optimize cloud spending across teams and environments:



AWS Cost Explorer

Visualizes and analyzes cloud spending trends across AWS accounts and services



ProsperOps

Automates commitment-based savings (like Reserved Instances) for AWS workloads

FinOps Tools

Below are the tools used in FinOps to monitor, analyze, and optimize cloud spending across teams and environments:



CloudZero

Tracks unit cost and allocates cloud spend to teams, products, or customers



Harness

Provides cloud cost visibility and integrates with CI/CD for cost-aware engineering

Quick Check



A company’s cloud spending is growing rapidly. They want developers to monitor usage and make cost-conscious decisions while maintaining delivery speed. What should they implement?

A. Cloud bursting

B. FinOps practices

C. ITIL ticketing

D. Cloud migration

Assisted Practice



Creating a Lambda Function to Launch a Virtual Machine **Duration: 15 Min.**

Problem statement:
You have been assigned a task to create an AWS Lambda function that automatically launches a virtual machine (EC2 instance). This includes configuring the IAM execution role, writing Lambda code in Python, and validating the instance launch.

Outcome:
By the end of this demo, you will be able to configure IAM roles and use AWS Lambda to programmatically launch EC2 instances, thereby automating infrastructure provisioning on AWS.

Note: Refer to the demo document for detailed steps:
01_Creating_a_Lambda_Function_to_Launch_a_Virtual_Machine

Assisted Practice: Guidelines



Steps to be followed:

1. Create the Lambda function to launch a virtual machine (VM)
2. Reconfigure IAM role to add EC2 permissions
3. Execute and test the Lambda function

ASSISTED PRACTICE

Key Takeaways

- The global cloud market is expanding rapidly, driven by digital transformation, hybrid workforces, and AI integration.
- Hybrid and multi-cloud strategies are becoming essential for flexibility, workload portability, and vendor risk management.
- Edge computing supports low-latency processing and real-time decision-making by moving computation closer to the data source.
- Sustainable cloud initiatives aim to reduce carbon emissions, with providers like Google and Microsoft leading 24/7 renewable efforts.



Key Takeaways

- AI-as-a-Service (AIaaS) enables organizations to use advanced AI capabilities via cloud-based APIs and platforms without building models from scratch.
- FinOps (Cloud Financial Management) helps organizations optimize cloud spending by promoting shared accountability and real-time cost visibility.
- Quantum computing in the cloud is making advanced research accessible, with early applications in simulation, cryptography, and optimization.



Thank You